NOVA IMS
INFORMATION MANAGEMENT SCHOOL

Big Data Analysis Project Handout 2026/2027

1. Project Objective

The objective of this project is to develop and present a complete Big Data Analytics project using **Apache Spark and Databricks**. You will work with real-world datasets and apply the concepts covered throughout the course to address a meaningful analytical problem, following an end-to-end workflow from data preparation and exploration to advanced analytics, graph analysis, and communication of results.

For this project, you and your group will act as a **data analytics consulting team at BigDataCompany**, a consulting firm that helps organizations work with large and complex datasets. Your team has been asked to develop an initial prototype for an imaginary client that wants to improve the way it processes and analyses its data and is considering **Apache Spark and PySpark** as the core technologies for its solution.

Your goal is therefore not only to produce interesting analytical results, but also to demonstrate **how Spark can support the different stages of the project** and why it is a suitable technology in a Big Data context. Throughout the project, you should make thoughtful use of Spark's capabilities for data processing, analytics, machine learning, and graph analysis. Working in **groups of 4 to 5 students**, your team will develop and execute the project using **Databricks Free Edition** as the main environment.

2. Relevant dates

- **11th December 2026 23:59:** Submission of the final project.

3. Project requirements and evaluation criteria

Each group must select **one of the three project topics** listed below, each of which has its own associated dataset:

1. **Text Mining on Mental Health**
2. **Asian Restaurant Sales**
3. **Churn Prediction**

Detailed information about each dataset, including its characteristics and potential applications, is provided in the **metadata.pdf** file available on Moodle.

The goal is to use the selected data to identify **interesting and meaningful insights** and, where appropriate, develop useful analytical tools such as forecasting, classification, clustering, or other data-driven solutions. You should demonstrate as broadly as possible the skills developed throughout the course, including data preparation, exploration, visualisation, data engineering, machine learning, and graph analytics. The project should not be a collection of isolated tasks. Your group should develop a **coherent analytical solution** in which the data, questions, methods, and results are clearly connected.

3.1. Required components

Your project should demonstrate your ability to work with **PySpark** and to understand how Spark can be used to address Big Data problems. At a minimum, you should address the following components:

1) **DataFrames & SparkSQL:** Use Spark DataFrames and/or Spark SQL to load, transform, query, aggregate, and analyse the data.
2) **Data Engineering & Pipelines:** Demonstrate an appropriate data preparation workflow, including tasks such as data cleaning, transformation, integration, and feature preparation.
3) **Machine Learning – Spark MLlib:** Develop at least one machine learning model or advanced analytics solution using Spark MLlib, where appropriate to your dataset and research/business question.
4) **GraphFrames:** Use GraphFrames to model and analyse relationships between entities, identifying relevant graph structures, patterns, or insights.
5) **Code Quality and Documentation:** Your notebooks should contain readable, well-structured, and appropriately documented code. Use meaningful variable and function names, avoid unnecessary repetition, and clearly explain important decisions made throughout the analysis.

Additionally, from the Spark code you are expected to deliver a PowerPoint presentation, aimed at management, meaning a non-technical audience. It should communicate the value of your work for the client: the insights you identified, the prototype tools you developed, and why Spark is an appropriate solution for the client's scale and needs. The presentation should not be technical. It should be framed as a consulting update, for example: "Here is how the client can make use of big data with Spark for these datasets and business questions," rather than: "Here are the model parameters," or "We used algorithm Y with hyperparameter settings Z." Keep the focus on Spark's role, the nature of the data challenges (think about the 4 V's), and what the client gains by using Spark. It must use the **official NOVA IMS PowerPoint template** and contain **no more than 10 slides**, excluding the title and references.

Online presentations and discussion sessions will take place in the **week after the submission deadline**. Each group must register on Moodle to receive a presentation slot; groups that do not register may not be given the opportunity to present and may face a **grade deduction**. During the session, groups will present their work and answer questions about their notebooks and modelling choices, so all members should be able to clearly explain their own work. Each group will have a **15-minute slot**, including **5 minutes for the presentation and 10 minutes for questions**. The grade will primarily reflect your **understanding and correct use of Spark**, so demonstrate relevant Spark functionalities while keeping the work clear, well-organised, and showing both **technical and managerial understanding of big data**.

3.2. Evaluation criteria

The project will be evaluated based on both the **technical quality of the Spark implementation** and the ability to **communicate the results and their value from a managerial perspective**. The final grade will be divided as follows:

- Technical implementation and Spark usage **(70%)**
- Presentation and discussion **(30%)**

4. Delivery guide

The final project must be submitted through Moodle by **11th December 2026 at 23:59** as a single ZIP file, named **GX_project.zip** (e.g., GD_project.zip). The ZIP file must contain **only the following deliverables**:

- **Executed PySpark Jupyter Notebooks**, containing the complete Python code, analysis, and **visible outputs**. The notebooks should be named **GX_something.ipynb** (e.g., GD_EDA.ipynb).
  - This is the **only code that will be considered for grading**.
  - You may also include .py files containing classes, functions, or other objects used in your project.
  - **Notebooks without visible outputs will not be graded.**
- **Final PowerPoint presentation exported as a PDF**, named **GX_presentation.pdf** (e.g., GD_presentation.pdf).

All submitted materials should be clearly organised and provide sufficient information to understand and reproduce the analysis.

Note: All deliverables must be submitted through Moodle. Late submissions of the final project will result in a penalty of **1.5 points per day** for each day past the deadline.

5. Bonus

In addition to the required project components, students may earn bonus points by demonstrating additional work that goes beyond the core requirements of the course. Although the final grade **cannot exceed 20/20**, bonus points can help students reach the maximum grade. There are **2 ways to earn bonus points:**

- **Spark Streaming (max. 1 point)** – Students may earn up to **1 additional point** by implementing a meaningful Spark Streaming component in their project. The streaming solution should be appropriately integrated with the project's datasets or business/research question and demonstrate how Spark can be used to process data continuously or in a streaming context. Simply including streaming functionality without a clear purpose or meaningful analysis will not be sufficient for the bonus.

The objective of this bonus point is to reward students who go beyond the project requirements and demonstrate additional effort, initiative, and commitment. While it is possible to achieve the maximum grade without bonuses, taking advantage of this bonus points increases the likelihood of reaching the highest mark. Each group should choose the approach that best suits its working style.